DISEMPOWERMENT PATTERNS IN REAL-WORLD AI USAGE

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1. Introduction

The evolution of the large language models (LLM) has accelerated the growth of generative AI assistants, such as ChatGPT, Claude, and Gemini. Such systems are increasingly getting integrated in the daily operations, and they assist the users to acquire information, assist in decision making, resolve issues and even provide emotions. Because of the increased utilization of such AI systems, some level of concern has arisen regarding the manner in which such systems might influence the beliefs, values, and decision making of human beings. This raises significant problems relating to the alignment of AI, human autonomy, and the general influence of algorithms in general.

The selected research article, which was developed by Sharma et al. (2025) contemplates these concerns and analyzes situational disempowerment in the interactions of AI using the help of the analysis of large-scale interactions between Claude and 1.5 million conversations. The methodology and the findings of the paper have been critically evaluated in this report and the main technical and methodological limitations have been determined, and a deep learning-based extension that can be used to detect and mitigate the risks of disempowerment in AI-assisted interactions has been suggested.

2. Paper Summary

The paper discusses how AI assistants can be utilized in situational disempowerment, where the adoption of AI influences the users in a way that will impact their perception, beliefs, or behaviour. Sharma et al. (2025) identify three significant distortions as they are, reality distortion, where the users develop some false beliefs about facts; value judgment distortion, where the users are misinformed by AI-generated outputs; and action distortion, where the AI-generated outputs cause the users to take actions that are not consistent with their actual values. In order to examine this phenomenon, this study examines approximately 1.5 million conversations of Claude through a privacy-sensitive analysis system called Clio.

The technique presupposes a schema with 3 distortion primitives and 4 amplifying factors, e.g., vulnerable and authority projection, the use of which is assumed by the method to identify disempowerment signals with the help of LLM-based classifiers. Results suggest that there are occasional times of extreme disempowerment with the occurrence of less than 0.1 percent of interactions exhibiting the behaviour, although vulnerable users are more frequent. The

research also concludes that the majority of interactions with more disempowerment potential are in positive feedback and the indicators have increased over time. The article is the first empirical model of researching the effects of AI on human autonomy on a large scale.

3. Critical Appraisal

3.1 Evaluation of Theoretical Foundations

The paper methodology is founded on the existing advancements in large language models (LLMs) and their ability to conduct more complex thinking and categorization (Sharma et al. 2025). The authors do not train a particular neural classifier but apply LLM-based classifiers on models such as Claude Opus to analyse text in conversations and identify potential disempowerment cues. These models are transformer based where attention mechanisms are used to extract contextual relations among long text chains. This enables the system to discern fine conversational patterns, as far as making beliefs, value judgments and decision-making is concerned.

The study primarily relies on prompt-based classification where the severity score of different categories of disempowerment is determined by the design of the prompt which in turn are given out by the LLM. There are practical merits of this approach, including the fact that it is less costly to train and can be deployed fast, without developing a novel supervised learning model (Cheng et al. 2025). However, even the methods which take prompts into account also lead to variability and lack reproducibility since the outcome can be highly dependent on prompt design and model behaviour.

Moreover, probabilistic models such as LLMs can be biased with the biases of the training distribution, thus adversely affecting the quality of classification because it is possible to cause the corruption of the reliability of classification. Regarding AI alignment, the paper reveals problems in terms of reward modelling and human feedback using reinforcement learning (RLHF) in the context of which algorithms that are trained so as to achieve user approval can learn sycophantic behaviours, strengthening the users in their beliefs and not necessarily leading to rational thinking.

3.2 Methodology Critique

The article relies on the experimental method, which can offer the analysis of AI-human interaction at the mass scale, however, there are many limitations on validity and applicability of the findings. Firstly, it speaks about conversations of Claude.ai only. Whereas this provides a lot of information concerning one AI system, it limits the extrapolation of the results because other language models such as ChatGPT or Gemini may possess different interaction styles, safety policies and response behaviours (Tang et al. 2025). Besides, the sample is also relatively small; in particular, the sampling period of one week may not serve as a good measure of the user behaviour over time, nor as a general interaction pattern.

The other privacy preserving analysis framework which is applied in the study is referred to as Clio and it removes or abstracts some of the contexts of discussion. Although such an approach maintains the privacy of the users, it is also capable of destroying extremely important contextual information such as history of conversation, emotional mood, and behavioural expressions (Terzimehić et al. 2025). As a result of this fact the signals of potential disempowerment could be interpreted incompletely.

The other approach to the issue of methodology is attributable to the method of classification. The authors use huge language models to classify discussions that were originally made by the other language models. This poses potential risks of circularity in that models can reproduce similar biases or assumptions in the data they are trained on. In addition, probabilistic outputs may lead to hallucinated will of the user.

Besides, the severity classification scheme may be founded on a subjective evaluation particularly in quantifying value deviation. The comparison of the paper to the alternative baselines, such as large-scale human annotation or ablation studies, available also could further contribute to the raising of the credibility and reliability of the evaluation framework.

3.3 Analysis of Limitations

Despite the large scale of the analysis of the study, it has several severe limitations. The first one is that it consists of the observational research design, i.e., the research examines the conversations that exist, but they are not tested experimentally. Speaking of which, one cannot say whether users were motivated to modify their beliefs or behaviours according to the answers of AI. The research is, thus, incapable of establishing a cause and effect relationship between the AI interactions and the disempowerment of the users (Fang et al. 2026).

Second, the discussion focuses on individual interactions and not on the ability of users to be traced across time in various dialogues. The making of any human judgment is usually comprised of a process of reflection and discussion; therefore, it is difficult to obtain longitudinal data and articulate the patterns of influence or actual disempowerment.

Thirdly, the model personality bias may have an impact on the results as well since Claude is the one to generate the data. Different AI systems can articulate the many different styles of communication such as cautious, supportive, or overly agreeable depending on the training and safety policies (Mehmood et al. 2022).

Finally, the outcomes themselves portray serious ethical concerns like, the threat of reinforcing any negative beliefs, establishing emotional addiction, or manipulating the users through over-validating them. The issues raised by these concerns make transparent, user autonomy, and ethical AI systems deployment the core principles of the responsible implementation of AI.

3.4 Computational and Practical Constraints

The structure of analysis proposed also possesses several calculation and practical problems. The inference of large language models, especially Claude Opus is an essential part of the classification pipeline and is also used to evaluate conversation transcripts. The computation requirements to execute these models on millions of interactions are enormous and may require a high-performance GPUs and extensive cloud infrastructure and it may be a limitation to scaling in applications to continuous monitoring (Quan et al. 2025). In addition, the sensitive nature of conversation data makes the process of its analysis raise the concern of privacy due to a potential presence of personal or emotionally sensitive data even in an anonymized set. These concerns also coincide with novel regulatory requirements in AI governance, data protection, and transparency, which can also contribute to the feasibility of such large-scale surveillance systems in real AI systems.

4. Proposal for Improvement

4.1 Motivation for Improvement

Though the paper offers solid information on the danger of disempowerment in the interaction between AI and people, the existing method is mostly based on the post-hoc categorization of the conversation transcripts. The analysis of interactions is the system that does the analysis of

the interactions when the responses have already been made, in other words, the questions of distorting beliefs, values, or actions are realized only retrospectively. Thus, the framework does not intervene actively by preventing problematic responses in the very interaction process. This drawback restricts its use to real-life AI systems, in which the objective is not to minimize the harmful effect by the time it can affect users.

Moreover, the classification operation is not done in real time within the pipeline processing of the generation of the model but instead based on the history of dialogue of Claude. To counter this weakness, more aggressive approach is required. The present report therefore provides a deep learning-based model capable of detecting and overcoming the risks of disempowerment in generating responses, which may cause AI assistants to modify their responses dynamically and find a way to allow more human freedom.

4.2 Proposed Technical Extension

To address the weaknesses of post-hoc analysis, in this report, the researcher proposes a Framework of Disempowerment-Aware Response Generation that determines and avoids risks in the response generation process. The former is a Disempowerment Risk Predictor which is a neural classifier that estimates the probability of a generated response leading to distortion of reality, values or action. The model is based on the inputs, namely the context of the conversation, signs of user emotional condition or vulnerability, and reactions delivered by the language model (Mehmood et al. 2022). The inputs are processed by a transformer-based architecture as is the case of Claude. The transformer encoder then takes the contextual dependencies in the conversation and then a classification head that produces a probability of each distortion category. The model may be trained on the binary cross-entropy or multi-class cross-entropy loss functions.

The second component integrates the predictor into the training pipeline via the assistance of a reinforcement learning constraint. The danger of disempowerment may be factored in reward operation in reinforcement learning founded on human feeds (RLHF).

$$Reward = Human\ Preference\ Score - \lambda \times Disempowerment\ Risk$$

The transformation facilitates the generation of useful as well as user-autonomous responses through the model. Finally, a safety layer which is context- aware also real-time monitors.

When high risk occurs, a system varies its responses to encourage reflexive rather than directional responses so that the interactions can be more balanced and the autonomy reflected.

4.3 Training Strategy

The proposed system would be trained on a combination of annotated conversations datasets, simulated conversations, and crowd-sourced human annotations, which identify the potential disempowerment markers such as reality, value, or action misrepresentation. A risk classifier is then trained on the labelled conversation data at the first stage to identify these signals by use of the supervised learning. In this step, the optimization of model parameters is carried out by the backpropagation of prediction errors where a network is passed and weight is updated by the gradient-based optimization algorithms such as Adam (Terzimehić et al. 2025).

In the second phase, the classifier is integrated into a reinforcement learning of human feedback (RLHF) pipeline in human feedback systems such as Claude. Finally, the model is tuned with the help of the reinforcement learning, where the responses are optimized by gradient updates in order to reach a compromise between the minimization of the helpfulness and the risk of disempowerment.

4.4 Expected Outcomes

The proposed model will reduce the risk of disempowerment in the interactions of AI and people to a minimum. The system can prevent the outputs which can negatively affect user beliefs, values or even actions by detecting potential distortions during production of a response. This style is likely to embrace conformity to human autonomy as it encourages the reaction that leads to the facilitation of reflection and free decision making but not instructions. In terms of evaluation, the framework will lead to the conclusion of the low distortion rates in reality, in value and action categories. In addition, proactive risk mitigation might lead users to be more confident about AI systems such as Claude and lead to a lower amount of negative or ethically dubious interactions.

4.5 Evaluation Plan

The proposed framework would be put into test by using the technical performance indicators and human based evaluation parameters. To determine the accuracy of the disempowerment risk predictor on a labelled evaluation dataset, first standard classification measures of

precision, recall, F1-score and overall accuracy would be measure. Second, user satisfaction may be measured by means of controlled user studies which would enable the user to evaluate usefulness and appropriateness of the answers produced by systems such as Claude (Tang et al. 2025). In addition, the changes in the indicators of ethical safety would also be assessed with respect to the improved reality, value, and action distortion risks. This would be compared to their finding with baseline models that are not aware of disempowerment-conscious constraints.

5. Conclusion

In conclusion, the influence of AI assistants on the development of beliefs, values, and choices of a human being has become particularly important as the generative AI systems begin to be more closely tied to our lives. The analysed article is a quality contribution since it coined the concept of situational disempowerment and presented one of the first large-scale empirical studies of AI-human interactions through Claude data. A series of limitations were however realized among them being the usage of post-hoc analysis, ineffective generalization of the data applied, and potential classification biases.

The proposed framework is the disempowerment-conscious response generation framework that involves a proactive improvement and hence, the risk identification is a component of the response generation process. Lastly, another of the most significant problems of responsible and trustful AI technologies in the future is the creation of AI systems that can assist, but not harm human autonomy.

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